

Vehicle Type Classification Using Fine-tuned Resnet-50 CNN for Heavy Vehicles

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Abstract—Heavy vehicle recognition is one of the vehicle type recognitions (VTR) that plays an important role in intelligent transportation systems, logistics optimization, and infrastructure management. However, the VTR implementation remains challenging due to visual similarities between vehicles in the heavy vehicle classes, and the limited availability of specialized datasets. This paper presents a deep learning-based framework for heavy vehicle type recognition utilizing the ResNet50 Convolutional Neural Network (CNN) architecture, and self-obtained dataset based on standard heavy vehicles in Malaysia. The system is specifically designed to classify side-view images of heavy vehicles into three classes: Class 3 Lorries (2 axles), Class 4 Lorries (3 axles and above), and Class 5 Buses, following Jabatan Kerja Raya Malaysia (JKR) vehicle classification standards. The methodology encompasses data acquisition from local roads, preprocessing techniques involving aspect ratio maintenance and normalization, feature extraction using ResNet50, and multi-class classification with a Softmax layer. Experimental evaluations demonstrate that the proposed model achieves an average of over 93% accuracy. Despite some misclassification between lorry subclasses, the system exhibits strong potential for practical deployment in traffic monitoring and automated toll collection. Future work will focus on expanding the dataset, optimizing the model for real-time inference, and integrating additional modalities to further enhance recognition performance.

Keywords—heavy vehicle recognition, convolutional neural network (cnn), resnet50, side-view image classification, intelligent transportation systems, deep learning, vehicle type recognition

I. INTRODUCTION

Recognizing and classifying heavy vehicles like trucks and buses is crucial for transportation planning, logistics, and traffic management, but it's a challenging task due to factors like inter-class similarity, intra-class variance, and environmental conditions. Various ITS application, such as vehicle counting, vehicle type recognition (VTR), often misclassify when the vehicle structure is almost similar, and they rely on high-resolution images to extract the vehicle features vehicle counting [1]-[3] Recent advancements in computer vision and deep learning, particularly Convolutional Neural Networks (CNNs), offer promising alternatives due to their ability to learn hierarchical features and adapt to diverse visual environments. Prior research by Roecker et al. [4] and Sun et al. [5] explored the application of CNN and DBN, respectively, in vehicle and object

classification, revealing significant gains in precision and adaptability. However, these studies were limited to general vehicle categories and were constrained by insufficient dataset diversity and computational resources. Existing methods often struggle with poor image quality, varying camera angles, and the lack of comprehensive datasets for heavy vehicle classes, especially those adhering to national standards such as JKR Malaysia's classifications [6],[7]. This study addresses these gaps by proposing a CNN-based system using the ResNet50 architecture, fine-tuned for classifying heavy vehicles from side-view images, aiming to enhance generalization, efficiency, and accuracy in Intelligent Transportation Systems. The contributions of this paper are as follows:

- We propose a heavy vehicle recognition model using a fine-tuned ResNet50 CNN based on side-view images and JKR classification standards.
- We evaluate the model's performance using precision, recall, F1-score, and accuracy, highlighting its strengths and identifying areas for improvement.

The remainder of this paper is structured as follows: Section 2 reviews related work in deep learning-based vehicle recognition. Section 3 details the proposed methodology, including data acquisition, preprocessing, model architecture, and training procedures. Section 4 presents the experimental results and performance analysis. Finally, Section 5 concludes the paper with discussions on practical implications, limitations, and directions for future research.

II. RELATED WORKS

In recent years, deep learning has emerged as a powerful tool for addressing the limitations of vehicle recognition techniques, particularly in challenging environments where occlusions, varying lighting, and modified appearances are common. These Intelligent Transportation Systems often struggle under such conditions, prompting a shift toward visual recognition approaches using image-based deep learning frameworks.

Earlier vehicle classification efforts often relied on Support Vector Machines (SVM), with Chen et al. [8] demonstrating SVM's effectiveness using colour histograms for vehicle classification. However, SVMs struggle in complex, multiclass scenarios due to the reliance on handcrafted features, as highlighted by Qi et al. [9]. In

contrast, CNNs have become the dominant approach, offering automatic feature extraction and resilience to lighting and environmental variations, as shown by Roecker et al. [4] and Dong et al. [10]. Although real-time capable CNN architectures like YOLO and Faster R-CNN have advanced the field, challenges remain in distinguishing between visually similar heavy vehicle classes, calling for deeper and more sophisticated models. Distinguishing between distinctive features has proven to be a difficult task due to complex feature extraction, which often causes vehicle misclassification [11],[12].

ResNet50, introduced by He et al. [13], has become a key architecture for overcoming these challenges by leveraging residual learning to enable deeper network training without degradation. Its application to transportation domains has been further demonstrated by Kundu [14], affirming its suitability for complex classification tasks like heavy vehicle recognition. Although DBNs have shown promise in multi-class transportation scenarios [5], [15], their computational demands make them less practical for real-time deployment compared to CNNs.

Recent studies by Maungmai and Nuthong [16], as well as Shvai et al. [17], explored hybrid and ensemble learning strategies, combining color, depth, and multiple models to enhance classification robustness under challenging conditions like occlusion and poor weather. Collectively, these findings underline the necessity of deeper, more flexible architectures and multimodal integration to advance heavy vehicle recognition.

Despite the progress, many existing studies focus on general vehicle types rather than fine-grained heavy vehicle subclass differentiation, such as distinguishing Class 3 from Class 4 lorries. Most works emphasize front or top views, which do not effectively capture the structural features needed for heavy vehicle identification. Moreover, side-view classification under real-world conditions where it include varying angles, lighting, and modifications. It has received limited attention, even though it is crucial for practical deployment in highway monitoring and toll collection systems. Addressing these gaps, this study proposes a ResNet50-based CNN framework tailored specifically for side-view heavy vehicle classification according to Malaysia's JKR standards [7], aiming for streamlined deployment in structured environments such as toll booths and checkpoints.

III. METHODOLOGY

A. Overview

This study proposes a deep learning-based framework designed to recognize and classify heavy vehicles from side-view images into three specific classes, which are Class 3 Lorries (Heavy 2 Axles), Class 4 Lorries (Heavy 3 Axles and Above), and Class 5 Buses, following the classification guidelines of JKR Malaysia. The methodology is illustrated in figure 1. The complete experimental procedure, comprising dataset acquisition, data preprocessing, model design, training, and evaluation, is detailed in the following subsections to ensure full reproducibility and transparency.

B. Data Acquisition

The dataset for this study was constructed using two primary sources, which are self-captured images taken at highways, industrial zones, rest areas with a DSLR camera

under various environmental conditions, and public datasets such as ImageNet, ImageCV, COCO, and Kaggle. The focus on side-view images was strategic, as these views reveal critical features like axle count, vehicle length, and structural design, enhancing classification precision for real-world applications like toll systems and smart traffic monitoring. Then, each image was manually examined and annotated into these three heavy vehicle classes. The final dataset comprised 223 images, with 75 images each for Class 3 and Class 4 Lorries, and 73 images for Class 5 Buses. These images were divided into training (70%), validation (15%), and testing (15%) sets to ensure proportional class distribution and reduce sampling biases.

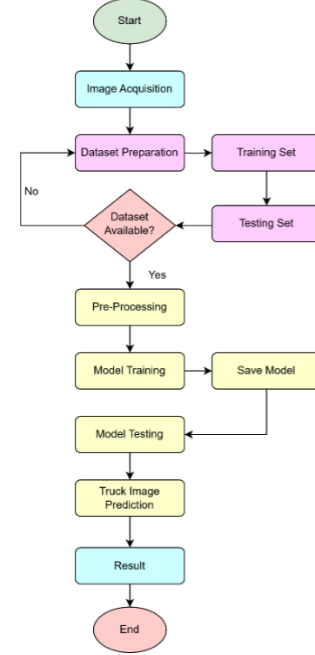


Fig. 1. Flowchart of the Proposed Heavy Vehicle Recognition Model

C. Data Pre-processing

All images underwent a standardized preprocessing pipeline to ensure consistency and facilitate effective model training. As shown in Figure 2, the first step was to resize the images to 224×224 pixels, which is the size that ResNet50 requires. While resizing, padding techniques were used to keep the aspect ratios the same so that the shapes of the vehicles didn't become distorted. This may have caused the loss of important visual signals needed for categorization.

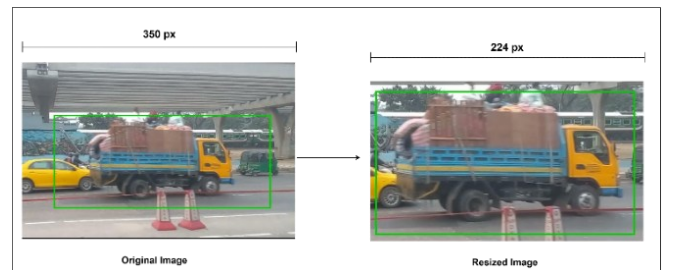


Fig. 2. Example of Data Pre-processing (Resizing)

After resizing, image normalization was done, which involved changing the pixel intensity values so that the mean was zero and the variance was one for each colour channel, as shown in (1):

$$X_{norm} = \frac{x - \mu}{\sigma} \quad (1)$$

where μ and σ are the mean and standard deviation computed over the training set.

This step was particularly crucial for keeping the numbers stable during network training, speeding up convergence, and minimizing the risks of gradients going missing or blowing up. We also employed data augmentation techniques like random flips, rotations, and zooms to add more variety to the scenarios. This makes sure that all the input data has the same scale and distribution, which helps the model learn.

D. Feature Extraction Using CNN and ResNet50

The ResNet50 deep Convolutional Neural Network (CNN) architecture is the main part of the proposed heavy vehicle recognition system. It was chosen because it is very good at deep residual learning, as shown in figure 3, which solves the vanishing gradient problem in deep networks. The model architecture starts with an input layer that accepts images resized to 224×224 pixels with three RGB color channels. This input dimension is specifically aligned with the original design of ResNet50 to leverage the benefits of transfer learning from the ImageNet pre-trained weights. Following the input, the network proceeds through a feature extraction stage consisting of multiple convolutional layers. Each convolutional layer is interspersed with Batch Normalization to stabilize the learning process and Rectified Linear Unit (ReLU) activation functions to introduce non-linearity, allowing the network to model complex feature relationships. Residual blocks are a key part of ResNet architectures. They group these convolutional layers together.

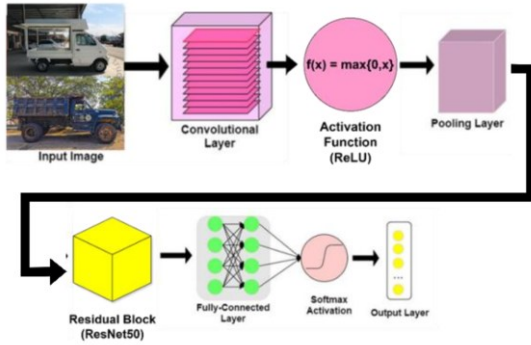


Fig. 3. Proposed Heavy Vehicle Recognition Model Architecture based on Resnet-50

These blocks have built-in skip connections or residual connections that let gradients flow directly between layers through identity mappings. This new idea makes it considerably easier to train deep models because it allows the network to learn residual functions instead of mappings that aren't used. This stops the network from getting worse as it goes deeper. A Global Average Pooling (GAP) layer takes the average of each feature map and merges them into one value after the features have been extracted. This cuts down on the number of parameters and the risk of overfitting. GAP also makes the connection between feature maps and output

categories stronger by introducing the spatially pooled features directly to the classification layer. This makes the model better at finding larger vehicles and more trustworthy.

E. Classification

The network has a fully connected (dense) layer that changes the taught features into a vector of class scores. After that, there is an output layer where a Softmax activation function makes a normalised probability distribution over the three types of heavy vehicles: Class 3 Lorries, Class 4 Lorries, and Class 5 Buses. The last guess is the class with the best chance. We made changes to the original ResNet50 architecture, which had been trained on ImageNet, so that it could be used for the heavy vehicle recognition challenge. We replaced the last fully connected layer with a custom dense layer that had three output nodes. Global average pooling before the dense layer reduces overfitting by minimizing the total number of parameters while maintaining effective feature summarization. Residual connections help efficient gradient flow, enabling the training of deeper networks without performance degradation. This architectural design uses the depth and robustness of ResNet50 while customizing its structure to suit the specific heavy vehicle recognition task, ensuring a good balance between model complexity, feature learning capacity, and generalization performance.

F. Training Strategies

We trained the model with the Adam optimizer and a learning rate of 0.001. We chose this optimizer because it can adjust the learning rates for different parameters, which helps the algorithm reach a solution faster. We chose a batch size of 32 since it was a decent balance between speed and memory restrictions. We used categorical cross-entropy as the loss function, which is great for problems with more than one class. The model was trained for up to 91 epochs, and it was stopped early based on validation loss to keep it from overfitting.

During training, data augmentation techniques such as random horizontal flipping, small rotations, and changes in brightness were used to make the dataset bigger and better at generalizing. We employed dropout regularization with a rate of 0.5 in fully connected layers to keep the model from fitting the data too closely. We checked the model when it was finished using metrics including accuracy, precision, recall, and F1-score. These are quite important for figuring out how well the model works, especially when it comes to telling the difference between trucks that look alike. We also made a confusion matrix to show how predictions are spread out and look for patterns of misclassification.

The heavy vehicle dataset is annotated with side-view photos, resized to 224×224 pixels, and normalized for replication purposes. A pre-trained ResNet50 model must be fine-tuned using a three-node Softmax output. Training must adhere to the specified hyperparameters, encompassing data augmentation, early termination, and dropout regularization. Evaluation must employ accuracy, precision, recall, F1-score, and confusion matrix analysis to guarantee alignment with the original data.

IV. RESULT AND DISCUSSION

A. Effect Proposal of a Heavy Vehicle Recognition Model using Fine-Tuned ResNet50 CNN

The main contribution of this study is a fine-tuned ResNet50 CNN to recognize heavy vehicle by training side-view images according to JKR classification standards. The results from the training and testing phases show that the fine-tuned ResNet50 CNN model is quite good for this job. It has high accuracy and generalizability, even with a little dataset.

The model was trained on a dataset with side-view images of heavy vehicles, including three categories: Class_3_Lorries, Class_4_Lorries, and Class_5_Buses. By referred to graph in figure 4, the training accuracy increased from 35% to 78%, and the validation accuracy showed a similar trend, indicating that the model was able to effectively generalize to discriminative features without overfitting. The consistent improvement in performance further supports the feasibility of using a fine-tuned ResNet50 for heavy vehicle classification based on side-view imagery.

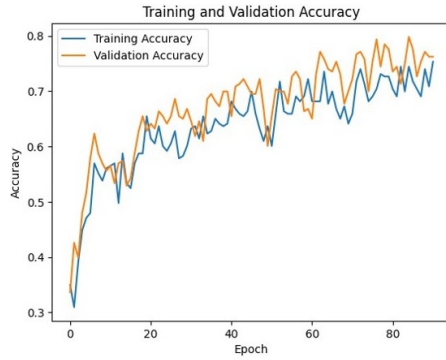


Fig. 4. Training and Validation Accuracy Graph

The deep learning approach employed in this study is more adaptable and precise than conventional methods such as SVM, which rely heavily on extensive manual feature extraction. A primary advantage of the ResNet50 model compared to its predecessors is its ability to independently learn hierarchical features from images without human intervention. This aligns with the objective of developing a vehicle recognition system that is more scalable and dependable, capable of effectively accommodating a diverse array of heavy truck kinds, as stipulated by the JKR classification requirements.

B. Evaluation of Model Performance Using Precision, Recall, F1-Score, and Accuracy

The second objective of this study is to assess the vehicle classification performance of the proposed heavy vehicle recognition model using standard evaluation metrics which are precision, recall, F1-score, and accuracy in table 1. Where the model achieved an overall accuracy of 78%, which also have been stated in earlier section.

TABLE I. COMPARISON VEHICLE CLASSIFICATION PERFORMANCE OF THE PROPOSED HEAVY VEHICLE RECOGNITION MODEL

	Class_3_Lorries	Class_4_Lorries	Class_5_Buses	Overall Average
Precision	0.81	0.63	0.84	0.78
Recall	0.70	0.85	0.85	0.78

F1-Score	0.84	0.85	0.84	0.77
Support	75	75	73	223

Looking at the results in table 1, the model performed exceptionally well in identifying Class_5_Buses, with high precision (0.84), recall (0.85), and F1-score (0.84), likely due to the distinctive visual characteristics of the buses, such as their elongated shape and size. Class_3_Lorries exhibited notable performance, attaining a precision of 0.81 and an F1-score of 0.84; nevertheless, the recall was comparatively lower at 0.70, signifying that some instances were missed. In contrast, Class_4_Lorries presented the most significant issue, exhibiting a precision of 0.63 despite a high recall of 0.85, which suggests a frequent misclassification of other vehicle types as Class_4_Lorries. These results highlight the model's strength in recognizing visually distinct vehicles while also pointing to difficulties in distinguishing between similar lorry types using only 2D side-view images where it shown the results in figure 5, and 6.



Fig. 5. Example of Testing Results in Class_3_Lorries



Fig. 6. Example of Testing Results in Class_4_Lorries



Fig. 7. Example of Testing Results in Class_5_Buses

TABLE II. CONFUSION MATRIX RESULTS OF THE PROPOSED HEAVY VEHICLE RECOGNITION MODEL

Actual Class	Predicted Class 3	Predicted Class 4	Predicted Class 5	Total (Support)
Class_3_Lorries	47	20	8	75
Class_4_Lorries	7	64	4	75
Class_5_Buses	4	7	62	73

To further understand these misclassifications, a confusion matrix analysis was conducted, as shown in table 2. This table showed that Class_3_Lorries were often mistaken for Class_4_Lorries, and sometimes the other way around. This makes sense because these two types of lorries can look very similar from the side. On the other hand, the model rarely got Class_5_Buses wrong, which shows it's quite good at recognizing them as shown in figure 7. In short, the model does a good job overall, but it has trouble telling apart vehicles that look alike. To make it better, future versions could use more camera angles or even 3D images to help the model see the differences more clearly.

C. Implications of findings

The real-world testing results show that the model performs well even in challenging conditions, such as different lighting, backgrounds, and vehicle appearances. It achieved high accuracy for buses (93.33%) and Class_4_Lorries (86.67%), however struggled more with Class_3_Lorries (70.00%), which confirms earlier findings. The results indicate that the model is reliable and feasible in practical environments such as toll booths and traffic monitoring systems. In comparison to traditional methods, our model, which utilizes a fine-tuned ResNet50, provides superior accuracy and demonstrates effective adaptability beyond the training environment.

The study also shows that using side-view images is a practical and cost-effective way to recognize heavy vehicles, especially in areas with limited access to advanced sensors. However, the model still has trouble telling apart vehicles that look very similar, like certain types of lorries. Future works may improve this by integrating supplementary data sources, such as rearward images, and broadening recognition capabilities to encompass other complex heavy vehicle types, including construction vehicles. These changes would enable the system's integration into smart city infrastructure, assisting in real-time traffic control, dynamic toll pricing, and enhanced road planning.

V. CONCLUSION

In this study, a deep learning approach was developed to recognize heavy vehicles using side-view images, focusing on Class 3 Lorries, Class 4 Lorries, and Class 5 Buses based on JKR Malaysia standards. By using a fine-tuned ResNet50 model and transfer learning, the system was able to handle challenges like limited data and the visual similarities between vehicle types. The model achieved 78% accuracy in testing, with especially strong results for bus class. These findings show that side-view images can provide enough visual information for identifying heavy vehicles, although it remains difficult to tell apart certain lorry types that look almost similar.

Overall, this research shows that deep learning models like ResNet50 can be adapted for real-world use without needing

expensive equipment. The method used here can be applied to other vehicle recognition tasks and could support systems like automated tolling and traffic monitoring. For future improvements, adding more types of data such as axle counts or sensor input, and using images from different angles could help the recognition model perform even better. Exploring lighter models could also make the system easier to use in real-time applications. This work provides a useful starting point for building smarter, more efficient transportation systems.

ACKNOWLEDGMENT

The authors would like to thank the Ministry of Higher Education for providing financial support under Universiti Malaysia Pahang Al-Sultan Abdullah for facilities, as well as additional financial support under Internal Research Grant Scheme (RDU243001).

REFERENCES

- [1] S. Awang, N. M. A. N. Azmi, and M. A. Rahman, "Vehicle Type Classification Using an Enhanced Sparse-Filtered Convolutional Neural Network With Layer-Skipping Strategy," *IEEE Access*, vol. 8, pp. 14265–14277, 2020.
- [2] S. Awang and N. M. A. Nik Azmi, "Automated Toll Collection System based on Vehicle Type Classification using Sparse-Filtered Convolutional Neural Networks with Layer-Skipping Strategy," *Journal of Physics: Conference Series*, vol. 1061, no. 1, p. 012009, 2018.
- [3] C. Xu et al., "Fast Vehicle and Pedestrian Detection Using Improved Mask R-CNN," *Mathematical Problems in Engineering*, vol. 2020, Art. no. 5761414, 2020.
- [4] M. N. Roecker, Y. M. G. Costa, J. L. R. Almeida, and G. H. G. Matsushita, "Automatic Vehicle Type Classification with Convolutional Neural Networks," in *Proc. 25th Int. Conf. Systems, Signals and Image Process. (IWSSIP)*, pp. 1–5, 2018.
- [5] N. Sun, G. Han, K. Du, J. Liu, and X. Li, "Person/Vehicle Classification Based on Deep Belief Networks," in *Proc. 10th Int. Conf. Natural Computation (ICNC)*, pp. 113–117, 2014.
- [6] Misman, N., & Awang, S. (2018). Camera-Based Vehicle Recognition Methods and Techniques: Systematic Literature Review. *Advanced Science Letters*, 24(10), 7623–7629. <https://doi.org/10.1166/asl.2018.12990>
- [7] JKR Malaysia, "Taklimat Bancian Trafik: Kursus Pengenalan Penilaian Kesan Trafik," Mohamad Wazir Shafiq bin Nordin, Public Works Department Malaysia (JKR), 2019. [Online]. Available: http://epsmg.jkr.gov.my/images/e/e7/Slide_Taklimat_Banc_i_Melaka.pdf
- [8] Z. Chen, N. Pears, M. Freeman, and J. Austin, "Road Vehicle Classification Using Support Vector Machines," in *Proc. IEEE Int. Conf. Intelligent Computing and Intelligent Systems (ICIS)*, pp. 214–218, 2009.
- [9] X. Qi, T. Wang, and J. Liu, "Comparison of Support Vector Machine and Softmax Classifiers in Computer Vision," in *Proc. 2nd Int. Conf. Mechanical, Control and Computer Engineering (ICMCCE)*, 2017.
- [10] W. Dong, Z. Yang, W. Ling, Y. Zhang, T. Liu, and X. Qiu, "Research on Vehicle Detection Algorithm Based on Convolutional Neural Network and Combining Color and Depth Images," in *Proc. 2nd Int. Conf. Information Systems and Computer Aided Education (ICISCAE)*, pp. 274–277, 2019.
- [11] Joodi, M. A., Al-Obaidi, F. E. M., & Al-Zuky, A. A. D. (2024). Improving vehicle classification accuracy with convolutional neural networks. 020015. <https://doi.org/10.1063/5.0229765>
- [12] Xu, C., Wang, G., Yan, S., Yu, J., Zhang, B., Dai, S., Li, Y., & Xu, L. (2020). Fast Vehicle and Pedestrian Detection Using Improved Mask R-CNN. *Mathematical Problems in Engineering*, 2020, 1–15. <https://doi.org/10.1155/2020/5761414>
- [13] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2016.
- [14] N. Kundu, "Exploring ResNet50: An In-Depth Look at the Model Architecture and Code Implementation," *IEEE*, 2023.

- [15] Y. Y. Wu and C. M. Tsai, "Pedestrian, Bike, Motorcycle, and Vehicle Classification via Deep Learning: Deep Belief Network and Small Training Set," in *Proc. Int. Conf. Applied System Innovation (ICASI)*, pp. 1–4, 2016.
- [16] W. Maungmai and C. Nuthong, "Vehicle Classification with Deep Learning," in *Proc. 4th IEEE Int. Conf. Computer and Communication Systems (ICCCS)*, pp. 294–298, 2019.
- [17] N. Shvai, A. Hasnat, A. Meicler, and A. Nakib, "Accurate Classification for Automatic Vehicle-Type Recognition Based on Ensemble Classifiers," *IEEE Trans. Intelligent Transportation Systems*, 2019.